SUPPLEMENTARY MATERIAL: TINYSHIP-FINGERPRINT

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ABSTRACT

Additional figures and tables for TinyShip-Fingerprint. The main paper reports the certified open-set protocol, Table I comparison, verification curves, enrollment grid, footprint, and OSCR. This supplement provides the CQT+MFCC front-end example, embedding t-SNE, comparison bar charts, FAR/FRR operating points, enrollment-sweep plots, and a short discussion.

Index Terms— underwater acoustics, open-set verification, supplementary material

1. FRONT-END

Fig. 1 shows the two-channel ShipNN-style front-end on a 1 s VTUAD clip.

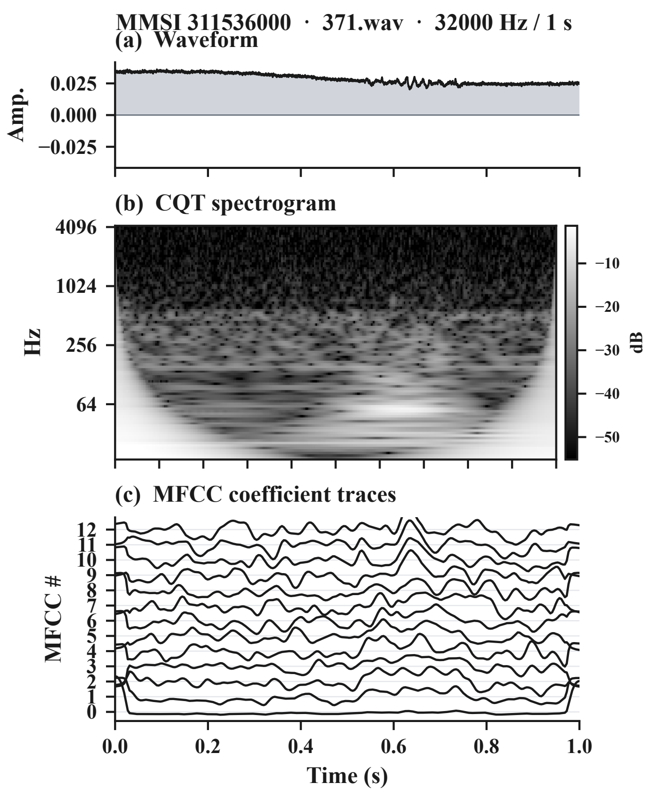

Fig. 1. Two-channel front-end on a 1 s VTUAD clip (MMSI 311536000, 32 kHz). (a) waveform; (b) CQT spectrogram (18–4186 Hz, log-magnitude); (c) 13 MFCC coefficient traces. CQT and MFCC are stacked as encoder input.

2. EMBEDDING SPACE

Fig. 2 visualizes 128-d embeddings of the 22 unseen evaluation vessels and MMSI=0 background clips.

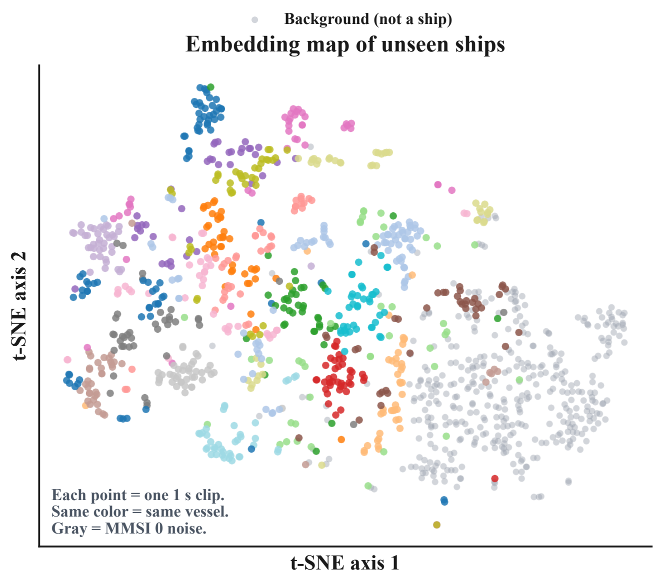

Fig. 2. t-SNE of 128-d embeddings on the 22 unseen test vessels (color = vessel) and MMSI=0 background clips (gray). Same-color clusters show identity grouping. Source: `fingerprint_attention_embeddings.npz`; t-SNE seed 42.

3. METHOD COMPARISON (BAR CHARTS)

Table I of the main paper lists exact mean±std. Fig. 3 is the corresponding bar-chart view.

4. FAR/FRR OPERATING POINTS

Table 1 complements the DET panel of main-paper Fig. 4 (seed 0).

5. ENROLLMENT / WINDOW CURVES

Table II of the main paper reports the $k \times w$ EER grid. Fig. 4 plots the same numbers.

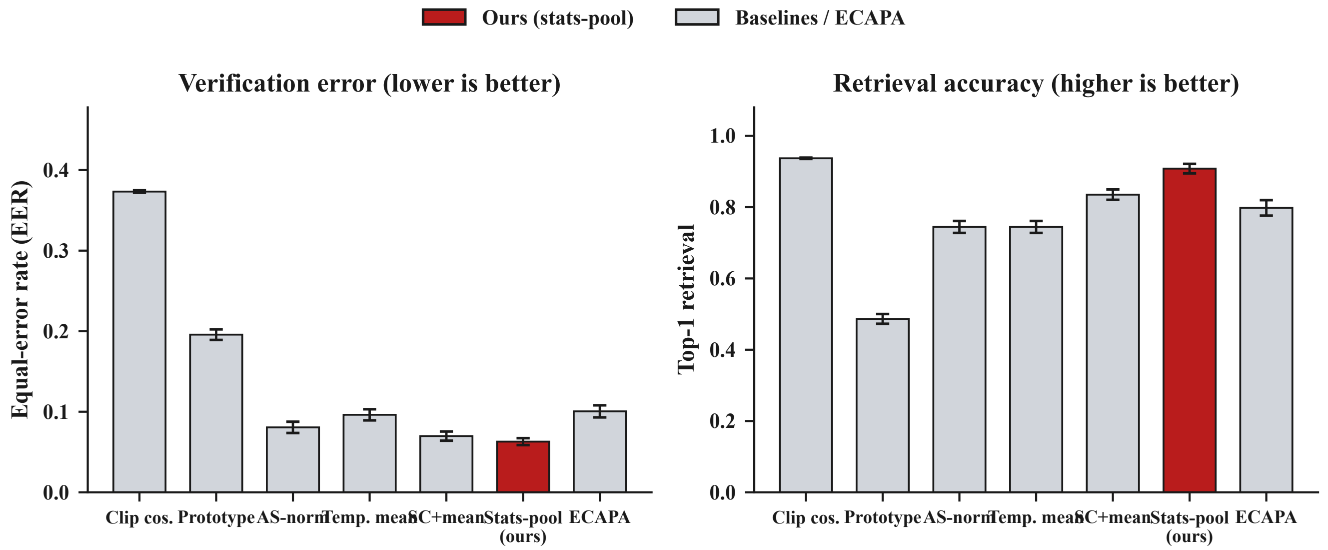

Fig. 3. Method comparison on the locked certified protocol (k=20, w=3, 22 vessels). Left: EER; right: top-1. Error bars are mean±std over seeds $\{0, 1, 2, 3, 4\}$ from `reports/table1_multiseed.json`. Accent color marks ours (stats-pool).

Table 1. FAR/FRR at seed 0 (k=20, w=3). Columns @FAR are FRR at that false-accept rate. ↓ = lower is better.

Method	EER↓	@.01↓	@.05↓	@.10↓
Mean pool	0.071	0.265	0.097	0.051
Stats-pool	**0.063**	**0.240**	**0.078**	**0.035**
ECAPA	0.114	0.503	0.229	0.130

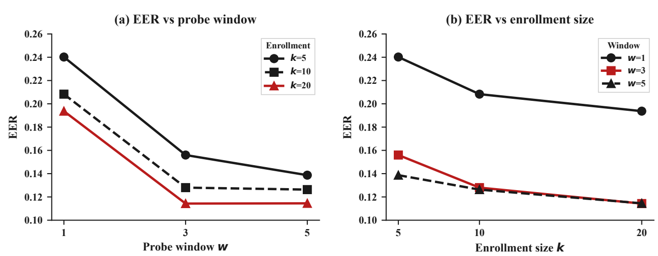

Fig. 4. Enrollment/window trade-off. Left: EER vs probe window w at fixed k. Right: EER vs enrollment k at fixed w. Data from `reports/free_experiments_arcface.json` (mean prototypes, seed 0, 22 ships).

6. OPEN-SET REJECTION

Table 2 reports CCR at FPR 0.1 and OSCR-AUC on the locked protocol.

7. DISCUSSION

We demonstrate the feasibility of open-set individual-vessel verification under MMSI-disjoint evaluation with a sub-million-parameter acoustic encoder (0.099 M). On the locked protocol, statistics-pooled prototypes deliver lower EER than larger speaker-verification-style ECAPA pipelines while retaining strong retrieval and open-set rejection, at ∼63× fewer parameters and a much smaller disk footprint. Enrollment and temporal window curves quantify the accuracy–enrollment cost trade-off. The distance-band audit in the main paper clarifies why we evaluate identity on MMSI-disjoint vessels rather than near→far closed-set transfer within VTUAD.

Table 2. Open-set rejection (w=3, k=20; 1297 unknown windows from 3891 background clips; mean±std, seeds $\{0, 1, 2, 3, 4\}$). ↑ = higher is better.

Method	CCR@0.1↑	OSCR-AUC↑
Sub-center + mean	0.530±0.044	0.717±0.024
Stats-pool (ours)	**0.597±0.037**	**0.794±0.010**
ECAPA-TDNN stats-pool	0.241±0.029	0.549±0.024